GeoRDF2Vec: exploratory analysis of dmg777k and BGS+

Semantic Networks project, Milestone 2

1. Goal and datasets

GeoRDF2Vec assigns a location to every node by flooding geometries through the graph, then biases RDF2Vec random walks towards geographically close neighbours. We plan two changes: relation-aware flooding and adaptive walk length. Before implementing them, we analysed what the data offers for each (`notebooks/dmg777k_eda.ipynb`, `notebooks/bgs_eda.ipynb`). Test labels were never used.

	dmg777k	**BGS+**
Graph	776,920 triples, 332,194 nodes, 60 relations	916,199 triples, 273,963 nodes, 103 relations
Task (majority)	5 classes, 8,396 labels (47.8%)	river (FLUV) vs glacial (GLACI), 146 labels
Geometries	21,208 WKT; every labelled node has its own	48,603 WKT; no labelled node has its own

We chose BGS+ as the second dataset (the paper's other one, DBpedia, was too big). It is a standard RDF2Vec/R-GCN benchmark, and differs from dmg777k in it's structure.

2. Structure, geometries and labels

	dmg777k	BGS+
Entities / literals	148,127 / 184,067	103,055 / 170,908
Degree	median 1; largest hub 65,576	median 1; largest hub 83,024
No outgoing edge	58.5% of nodes	69.5% of nodes
Relations	60; top 5 hold 42.5% of edges	103; top 5 hold 29.6% of edges

Both graphs are heavy-tailed multigraphs (56k and 95k node pairs with several relations) with one connected component, so flooding reaches every node. The big hubs are classes and shared values (`Not Entered`, photographers' names), not places. Most nodes are literals, which end any walk that follows edge direction.

- **Geometries.** dmg777k: 11,315 polygons and 9,893 points. 99.9% of monuments have 2–6 geometries, and 44,710 geometry nodes are empty stubs linked only to their class. BGS+: 23,917 polygons, 21,943 points and 2,743 lines, in British National Grid metres, which we reproject to WGS84 (36 invalid, 2 empty). Coverage is uneven: monuments cluster along rivers and in South Limburg; stratotypes where geology is well studied.
- **How locations reach the labelled nodes.** Each Dutch monument owns its geometry (2 hops: `hasGeometry` → `asWKT`). No BGS+ rock unit does. Only 3,290 of 17,943 units have a stratotype, and the map links to just 154 units, none of them labelled.
- **Labels.** dmg777k has 5 unnamed classes, with moderate imbalance (3.8% to 47.8%), so we report macro F1. BGS+ has 93 FLUV and 53 GLACI, we split it 94 / 23 / 29. One test unit moves accuracy by 3–4 points, so we use repeated cross-validation over training and validation units. 144 of the 146 BGS+ units are young superficial deposits, not bedrock.
- **Leakage.** The target relations are absent. In dmg777k, near-pure features (specific cities, streets) cover 1.3% of validation monuments. In BGS+, 8 features pure in training (parent unit, the age `QO`, texts like "Ground surface.") cover 9 of 23 validation units, 8 correctly. Graph structure alone explains much of the BGS+ class, consistent with published RDF2Vec accuracy of about 0.87.

3. Findings

Location relates to the class: modestly in dmg777k, strongly in BGS+. A nearest-neighbour classifier on coordinates alone reaches macro F1 0.26 on dmg777k (shuffled labels: 0.13). On BGS+, using a unit's stratotype it reaches 0.87 (shuffled: 0.47). Glacial deposits lie in the north (Figure 1, left).

Flooding gives many nodes a meaningless location. Flooding crosses every relation, including `rdf:type`. A class node linked to almost every geometry therefore inherits their national centroid and passes it on. In dmg777k this

affects 18.5% of nodes (via `geosparql:Geometry`). In BGS+ it affects 50.3% (via `skos:Concept`), including 49 of the 146 labelled units.

Relations differ sharply in how spatial they are. We grouped located entities by a shared object and measured each entity's median distance to its group's centre:

Kind	dmg777k (baseline 75 km)	BGS+ (baseline 248 km)
Spatial	postal code 30 m, city 370 m, municipality 2 km	location description 80 m, stratotype 4 km
Regional	province 22 km	parent unit 44 km, map units of one entry 85 km
Not spatial	build year, house number, `type`: 66–81 km	ages, colour, status, `type`: 106–248 km

Relation-aware flooding helps measurably on BGS+, before any training. We hid the stratotype links of the 97 labelled units that have one, re-ran flooding, and measured the distance to the true site:

Flooding variant	Labelled units at national centroid	Hold-out median error	Location-only macro F1
Repo (all relations)	49	389 km	0.72
Without `rdf:type`	0	72 km	0.76
Without hubs (degree > 1,000)	0	51 km	0.71
Without schema/attribute relations	0	**59 km**	**0.78**

Walk direction decides whether walk length matters (Figure 1, right). Walks that follow edge direction, as the authors' code does, end on a literal: with a limit of 8 they average 1.6–1.8 steps on both datasets, so the length limit is almost never reached. Walks that may also go backwards always reach 8 steps and keep drifting away from the start: 16 → 87 km on dmg777k, 156 → 257 km on BGS+. Geographic weights keep them 7–21 km (dmg777k) and 28–53 km (BGS+) closer. However, 91–97% of undirected BGS+ walks pass through a national-centroid hub.

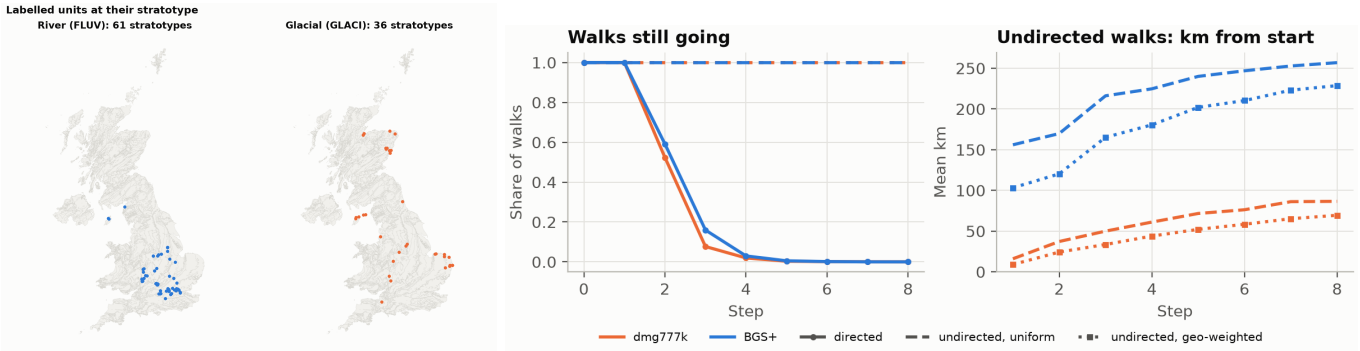


Figure 1. Left: BGS+ labelled units at their stratotype, by class. Right: random walks from labelled nodes (limit 8): share still going per step, and mean distance from the start for undirected walks, uniform vs geo-weighted.

4. Consequences for our contributions

Contribution	Evidence	Next step
Relation-aware flooding	Hubs corrupt 18.5% / 50.3% of locations. On BGS+, dropping non-spatial relations cuts hold-out error 389 → 59 km and raises F1 0.72 → 0.78. Dropping by degree does not raise F1.	Derive the relation list from measured dispersion. Run the hold-out on all 3,290 BGS stratotypes and on dmg777k.
Adaptive walk length	Directed walks end after about 1.7 steps. Undirected walks reach full length and keep drifting.	Fix walk direction first, then compare length rules on both datasets.

Limitations and baseline. The authors' code needs fixes before a baseline run (word2vec trains on no data as committed; longitude and latitude are swapped; see the notebooks). BGS+ has only 146 labels (fold spread ±0.08 macro F1), so we will compare methods with repeated cross-validation and paired tests. The kNN scores measure how much signal location carries, not embedding quality.